

System Identification

Class one

Basic Concepts and Overview of Linear Time Dynamic Systems

Bibliographical references :

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Van den Hof, P.M.J., 2020. System Identification - Data-Driven Modelling of Dynamic Systems, Lecture Notes, Department of Electrical Engineering, Eindhoven University of Technology, Eindhoven, The Netherlands.

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1 - Basic Concepts of System Identification

1.1 Systems

A system is a set of interconnected components that work together as a unified whole to achieve a specific purpose or function. Systems can be found in various contexts, Including:

- ▶ **Biological Systems:** In biology, a system might refer to something like the digestive system, where organs and processes work together to break down food and absorb nutrients.
- ▶ **Technological Systems:** In technology, a system could be a computer system, where hardware and software components interact to perform tasks and process information.
- ▶ **Ecological Systems:** In ecology, an ecosystem is a system where living organisms and their physical environment interact in a complex web of relationships.
- ▶ **Social Systems:** In social sciences, systems can refer to structures like economic systems, where various institutions and policies work together to manage resources and trade.
- ▶ **Mathematical and Scientific Systems:** In these fields, a system might refer to a set of equations or a model that describes relationships between variables or phenomena.

Essentially, a system is characterized by its organized structure and the way its components interact to maintain some kind of balance or achieve a specific outcome.

1.2 Models

Engineering models: They are concerned with relations between systems variables that are well-defined physical quantities, and moreover these quantities can very often be measured by some appropriate equipment.

One can measure forces, accelerations, temperatures, flows, voltages, currents, pressures etcetera, and this possibility of measuring provides the engineer with important information in the process of constructing and of using models.

- i) mental models. These reflect the intuitive notion that people have of a system behaviour. Take e.g. the human operator that is able to control a complicated system very well just by experience (captain of a huge tanker, people driving a car or bicycle, pilot steering a airplane).
- ii) software models. These reflect system descriptions that are contained in (extensive) software programs, possibly including if-then rules, discrete-event types of mechanisms and look-up tables (schedule of a railway company)
- iii) graphical models. System descriptions in the form of (nonlinear) characteristics and graphs, Bode and Nyquist plots, impulse and step responses.
- iv) mathematical models. System descriptions in the form of mathematical relations as e.g. (partial) differential and difference equations, fuzzy type of models and neural networks.

A model is determined by a number of (external) variables or signals that exhibit a specific relation with each other. There are different signals that we can distinguish:

- i) Measurable output signals. These are measurable signals that can be considered as consequences or responses of the system behaviour; they can not be manipulated directly by the experimenter.
- ii) Measurable input signals. These are measurable control signals that act as a cause of the system response. Generally they can be manipulated to a certain extent. If they can not be manipulated they are sometimes considered as (measurable) disturbance signals.
- iii) Non-measurable disturbances. These reflect non-measurable disturbances that act on the system. These disturbances can not be manipulated nor influenced.

1.3 Mathematical Models

Types of mathematical models:

- ▶ White box modeling. Modeling by the physics or nature of the process; phenomenological or conceptual modeling
- ▶ System identification. Alternative techniques to white box modeling (black box modeling). It is an experimental technique in which the fact that the system is in operation is used to, through measurements, build a model of the system.
- ▶ Gray box modeling (identification). When knowledge about the system is used that is not found in the data used for identification.

A mathematical model of a real system represents some observed features of this system, but not all.

A mathematical model is used for: forecasting; analysis and design of control systems; supervision, error detection and diagnosis; optimization.

The linear mathematical model has the same behavior regardless of the operating point; superposition principle.

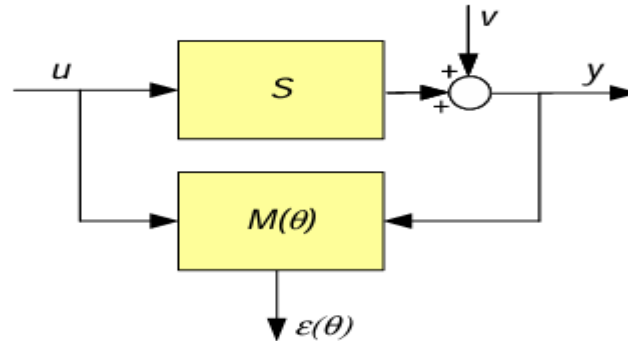
A time-invariant model has unchanging dynamics (the cause-and-effect relationship does not vary significantly over time).

A model with concentration of parameters has dynamics that depend only on time. If it also depends on another variable, such as distance, the model has non-concentrated parameters.

1.4 Identification procedure

Consider the situation as depicted in the block diagram in Fig 1.1. Here u and y reflect the input and output signal, respectively. Signal v acts as a nonmeasurable disturbance on the output. S is the system to be modelled. $M(\cdot)$ reflects a parametrized set of models, where the parameter Θ can range over a parameter set $\Theta \subset \mathbb{R}^d$. For each $\theta_1 \in \Theta$, $M(\theta_1)$ is a model that relates input and output signals to each other.

Fig. 1.1



However, when confronting a model $M(\theta_1)$ with measurement data (y,u) , the model relations will generally not be satisfied exactly, as noise influences on the data will circumvent this to happen. This implies that when confronting a model with measurement data, an error or residual signal will result. This is reflected in Fig. 1.1 by the residual signal $(\varepsilon(\theta_1))$.

1.4 Identification procedure

In a general identification problem one can consider the following principal choices to be made:

- *Experiment.* Measurements of input and output signals have to be gathered, leading to a data set $\{y(t), u(t)\}_{t=1, \dots, N}$. When designing an experiment, the characteristics of the input signal u have to be chosen so as to excite the relevant dynamics in the system S in order to get information on these relevant dynamics reflected in the measured output signal.
- *Model set.* Every model in the set can e.g. be represented by a difference equation

$$y(t) + a_1 y(t-1) + a_2 y(t-2) = b_1 u(t-1) + b_2 u(t-2) + \varepsilon(t) \quad (1.1)$$

with $t = 1, 2, \dots$, and where the parameter vector $\theta \in \mathbb{R}^4$ is composed as $\theta = (a_1 \ a_2 \ b_1 \ b_2)^T$. After having specified the number of parameters (and thus the order of the difference equation), a model set is obtained by varying θ over a parameter set $\Theta \subset \mathbb{R}^4$.

- *Identification criterion.* Given measurement data $\{y(t), u(t)\}_{t=1, \dots, N}$, each model (and therefore each parameter $\theta \in \Theta$) will generate a residual signal

$$\varepsilon(t, \theta) = y(t) + a_1 y(t-1) + a_2 y(t-2) - b_1 u(t-1) - b_2 u(t-2) \quad (1.2)$$

which is now dependent on the parameter θ .

1.4 Identification procedure

This residual signal $\varepsilon(t, \theta)$ can be used as a basis for constructing the identified model(s), and this can be done in several different ways:

The most standard and classical way is to construct a cost function, which is minimized over θ . The most popular choice is a least-squares criterion:

$$V_N(\theta) = \frac{1}{N} \sum_{t=1}^N \varepsilon^2(t, \theta)$$

and a parameter estimate is constructed by finding that value of θ that minimizes $V_N(\theta)$. This is written as:

$$\hat{\theta}_N = \arg \min_{\theta \in \Theta} V_N(\theta) \quad (1.3)$$

meaning that $\hat{\theta}_N$ is that value of θ that minimizes the sum-of-squares criterion.

However there are many alternatives to this approach, considering different types of identification criteria. In an instrumental variable (IV) criterion one selects the estimated parameter $\hat{\theta}_N$ so as to make $\varepsilon(t, \hat{\theta}_N)$ uncorrelated to some external (instrumental) signal, and in the approach of so-called *set membership identification*, the identification criterion delivers a set of estimated parameters $\hat{\theta}_N$ that satisfy $|\varepsilon(t, \hat{\theta}_N)| < c$ for some prechosen level c .

1.4 Identification procedure

The different aspects that are crucial in any identification procedure are reflected in a scheme in Figure 1.9, which is due to Ljung (1987).

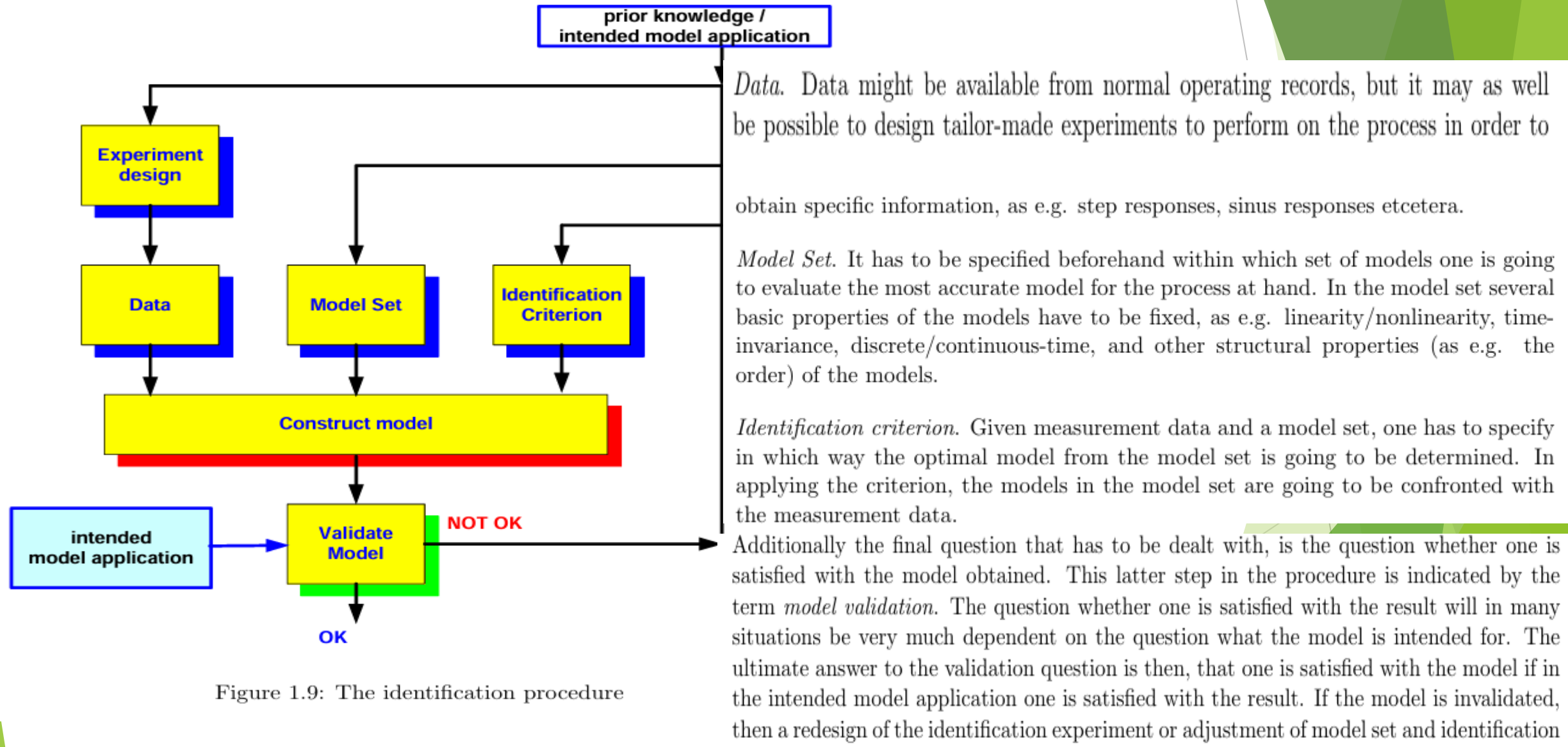
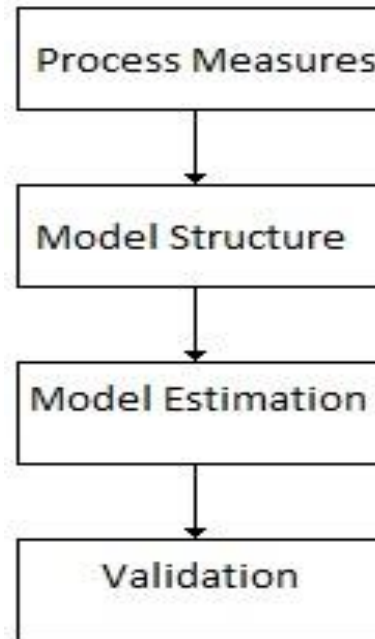


Figure 1.9: The identification procedure

1.4 Identification procedure

Aguirre and Coelho et al (2000), establish the following identification procedure for parametric models, which is analogous to that shown previously:



The model set and identification criteria are similar to Model Structure and Model Estimation, respectively.

1.5 Basic questions about System Identification

- ▶ What is System Identification? Building mathematical models of a dynamic system based on measured data.
- ▶ How is this done? By adjusting parameters of a model until its output matches the measured output as closely as possible.³
- ▶ How do you know if a model is good? A good test is to compare the model output with the measured data set that was not used for identification (data validation).
- ▶ The quality of the model can be tested in other ways? By analyzing what the model cannot reproduce from the data, the residuals
- ▶ What are the most common models? The most common are those described by difference equations, but other models such as neural networks and fuzzy models are also widely used.
- ▶ Does a specific model have to be chosen? For parametric models, it is necessary to specify the structure.
- ▶ How do you know which structure to try? Trial and error.

2 - Overview of Linear Dynamic Systems

2.1 Time Continuous and Linear Dynamical Systems

Systems can be represented by the transfer function, impulse response or through state equations. In the case of linear continuous systems, these representations are as follows:

i) Continuous Transfer Function: The transfer function of a linear system, defined by $H(s)$, is the ratio of the Laplace transform of the output, $Y(s)$, to the Laplace transform of the input, $U(s)$. This relation is defined by:

$$H(s) = \frac{B(s)}{A(s)} \quad \text{where} \quad B(s) = \sum_{j=0}^m b_j s^j \quad ; \quad A(s) = \sum_{i=0}^n a_i s^i$$

ii) Continuous Impulse Response: In a linear time-invariant system the input ($u(t)$) and output ($y(t)$) signals are related by the convolution integral given by:

$$y(t) = \int_0^t h(t-\tau) u(\tau) d\tau \quad \text{where } h(t) \text{ is the impulse response of the system given by: } h(t) = \mathcal{L}^{-1}[H(s)]$$

iii) Continuous Equations of State: The transfer function is related to the representation of states, in the case of monovariable states, as follows:

$$H(s) = c(sI - A)^{-1}b$$

where A is the system matrix ($n \times n$), b is the input vector ($n \times 1$) and c is the output vector ($1 \times n$). The state equations are

$$\dot{x}(t) = Ax(t) + bu(t)$$

$$y(t) = cx(t)$$

where $x(t)$ is the state vector ($n \times 1$), $u(t)$ is the input and $y(t)$ is the output.

2.2 Time Discrete and Linear Dynamical Systems

i) The discrete transfer function is the ratio of the z-transform of the output, $Y(z)$, to the z-transform of the input, $U(z)$, i.e., $H(z) = \frac{Y(z)}{U(z)}$

The relation $H(z)$ is a ratio of two polynomials in z and is represented by $H(z) = \frac{B^d(z)}{A^d(z)}$

where $B^d(z) = \sum_{j=0}^m b_j^d z^j$; $A^d(z) = \sum_{i=0}^n a_i^d z^i$

ii) The discrete impulsive response is related to the transfer function by: $h(t) = Z^{-1}[H(z)]$
where Z^{-1} is the inverse of the z-transform and t is the discrete time.

The impulse response samples, $h(t)$, are related to the step response samples, $s(t)$, by the following equations:

$$h(t) = \frac{1}{T_s} [s(t) - s(t-1)] \quad s(t) = \sum_{i=0}^t h(i)$$

iii) The transfer function is related to the discrete representation of states, in the monovariable case, by: $H(z) = c(zI - A^d)^{-1} b^d$

where A^d is the system matrix ($n \times n$), B^d is the input vector ($n \times 1$) and c is the output vector ($1 \times n$), of the following state equations in discrete form:

$$x(t+1) = A^d x(t) + b^d u(t)$$
$$y(t) = cx(t)$$

2.2.1 Discretização da FT de sistemas contínuos

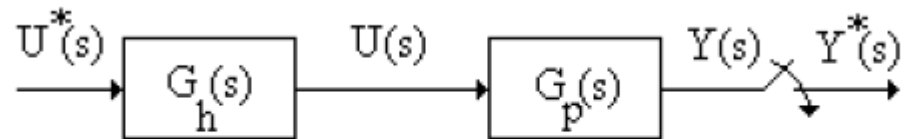


Figura 3.6 - Segurador de ordem zero em série com o processo.

onde $G_p(s)$ é a função de transferência do processo, $G_h(s)$ é a função de transferência do segurador de ordem zero, $Y(s)$ e $Y^*(s)$ são os sinais de saída contínuo e pulsado (discreto) e, $U(s)$ e $U^*(s)$ são os sinais de entrada contínuo e pulsado (discreto).

A função de transferência amostrada do sistema é calculada por

$$\frac{Y(z)}{U(z)} = Z\{G_h(s)G_p(s)\} = (1 - z^{-1})Z\left\{\frac{G_p(s)}{s}\right\} \quad (3.7)$$

e sendo $G_h(s)$ da forma

$$G_h(s) = \frac{(1 - e^{-sT_s})}{s} \quad (3.8)$$

2.2.1 Discretização da FT de sistemas contínuos

Exemplo 3.3 - Considere o processo de primeira ordem representado por

$$G_p(s) = \frac{K_1}{\tau s + 1}$$

onde $K_1 = 1$, $\tau = 7.5 \text{ seg}$, $T_s = 4 \text{ seg}$. Obter a função de transferência discreta sem e com o segurador de ordem zero.

$$G_p(s) = \frac{K_2}{s + a} \quad ; \quad a = 1/\tau \quad ; \quad K_2 = K_1/\tau$$

i) Caso sem o segurador de ordem zero

$$G_p^-(z) = \frac{b_0}{1 + a_1 z^{-1}} \quad ; \quad b_0 = K_2 = K_1/\tau = 0.133 \quad ; \quad a_1 = -e^{-aT_s} = -0.5866$$

ii) Caso com o segurador de ordem zero

$$G_p^+(z) = \frac{b_0 z^{-1}}{1 + a_1 z^{-1}} \quad ; \quad b_0 = (1 - e^{-aT_s}) K_2/\tau = 0.4134 \quad ; \quad a_1 = -e^{-aT_s} = -0.5866$$

Pode-se observar que $G_p^-(z) \neq G_p^+(z)$ e, assim, os casos (i) e (ii) conduzem a resultados onde tem-se modelos matemáticos diferentes (na estrutura e nos parâmetros).

2.3 Difference Equations

Let us consider the general model:

$$A(q)y(k) = \frac{B(q)}{F(q)}u(k) + \frac{C(q)}{D(q)}v(k),$$

where, q^{-1} is the delay operator such that $y(k) q^{-1} = y(k-1)$, $v(k)$ is a White noise, and $A(q), B(q), C(q), D(q), F(q)$ the polynomials defined as follows:

$$A(q) = 1 - a_1q^{-1} - \dots - a_{n_y}q^{-n_y};$$

$$B(q) = b_1q^{-1} + \dots + b_{n_u}q^{-n_u};$$

$$C(q) = 1 + c_1q^{-1} + \dots + c_{n_v}q^{-n_v};$$

$$D(q) = 1 + d_1q^{-1} + \dots + d_{n_d}q^{-n_d};$$

$$F(q) = 1 + f_1q^{-1} + \dots + f_{n_f}q^{-n_f}.$$

i) Autoregressive with and without input (ARX) and (AR), respectively:

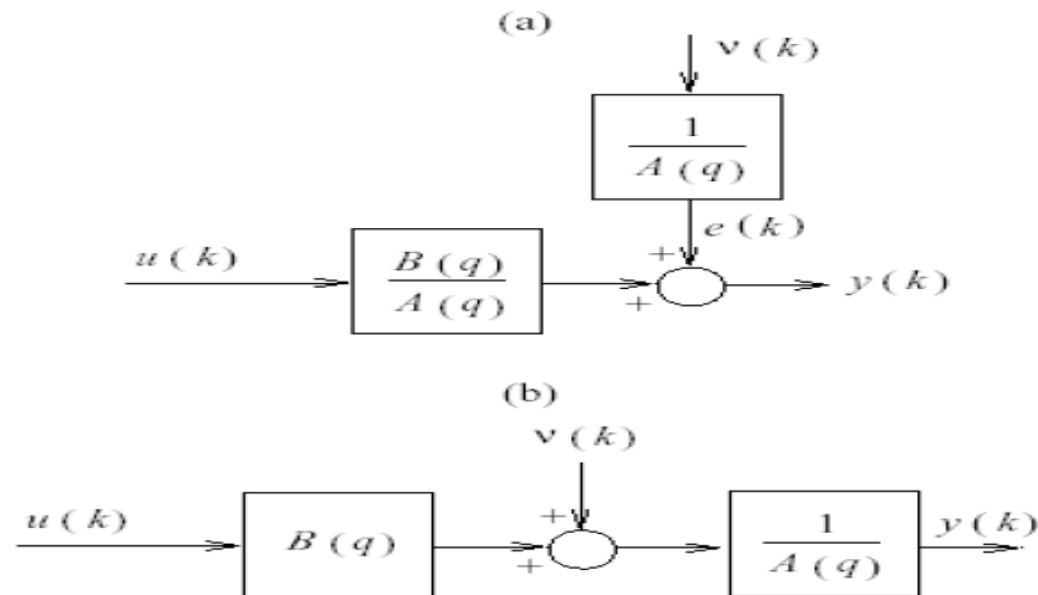
► AR: $A(q)y(k) = v(k)$, or equivalently:

$$y(k) = a_1 y(k-1) + a_2 y(k-2) + \dots + a_{n_y} y(k-n_y) + v(k)$$

► ARX: $A(q)v(k) = B(q)u(k) + v(k)$

Notice that $y(k) = \frac{B(q)}{A(q)}u(k) + \frac{1}{A(q)}v(k)$. This means that the output error signal $e(k) = \frac{v(k)}{A(q)}$

is not a white noise and this error represents a modeling error, i.e., it is affected by the polynomial $A(q)$. Next it is shown the block diagram of the ARX model



ii) Autoregressive model with moving average and exogenous input (ARMAX)

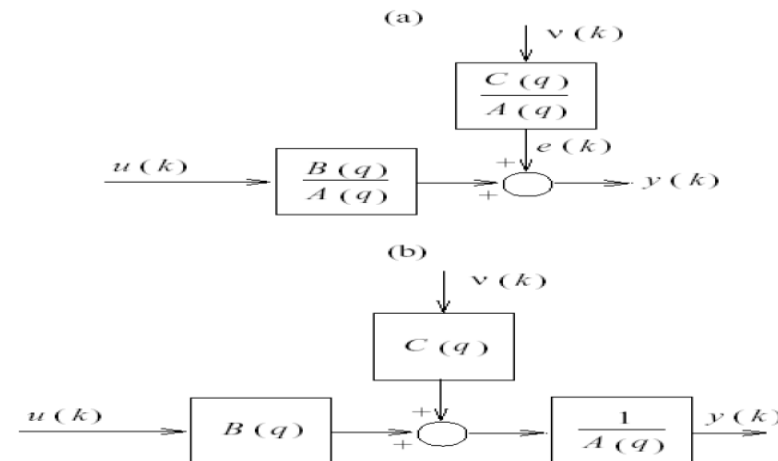
$$A(q)y(k) = B(q)u(k) + C(q)v(k),$$

$$y(k) = \frac{B(q)}{A(q)}u(k) + \frac{C(q)}{A(q)}v(k)$$

Which is equivalent to:

$$y(k) = H(q)u(k) + e(k),$$

where $e(k)$ is non-white. The ARMAX model belongs to the class of models in the equation. In the present case, the error in the equation is modeled as a moving average (MA) process, and the error signal added to the output, $e(k)$, is modeled as white noise $v(k)$ filtered by an ARMA filter C/A . See in next figure the block diagram of an ARMAX model:

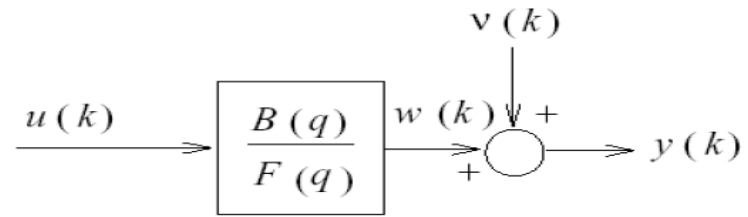


iii) Output error model:

$$y(k) = \frac{B(q)}{F(q)}u(k) + \nu(k),$$

It is common to represent this model using an auxiliary variable w :

$$\begin{aligned} F(q)w(k) &= B(q)u(k) \\ y(k) &= w(k) + \nu(k). \end{aligned}$$

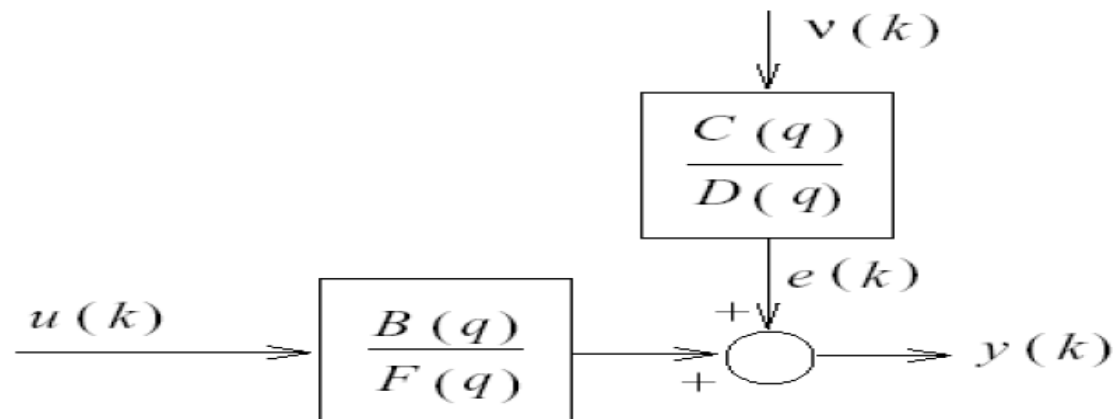


iv) Box and Jenkins model:

- ▶ This model can be determined from the general model, by taking $A(q) = 1$ and choosing arbitrarily the other polynomials, which result in the following expression:

$$y(k) = \frac{B(q)}{F(q)}u(k) + \frac{C(q)}{D(q)}v(k).$$

- ▶ The $B(q)/F(q)$ and $C(q)/D(q)$ transfer functions have no parameters in common (they are independently parameterized).
- ▶ This is also an output error model and its block diagram is drawn below:



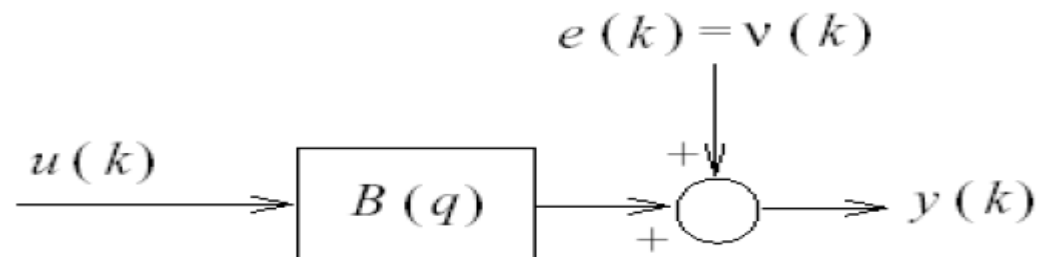
v) Finite Impulse Response (FIR)

$$y(k) = \sum_{j=0}^M h(j)u(k-j) + e(k).$$

- This model is equivalent to the general model, by taking $A(q) = C(q) = D(q) = F(q) = 1$ and $B(q)$ arbitrarily:

$$y(k) = B(q)u(k) + v(k).$$

- This is also an output error model, and its block diagram is:



2.3.1 Eq. a diferenças a partir da FT discreta

$$G_p(z) = z^{-d} \frac{B(z^{-1})}{A(z^{-1})} = \frac{Y(z)}{U(z)} \quad (3.13)$$

ou

$$A(z^{-1})y(t) = z^{-d}B(z^{-1})u(t) \quad (3.14)$$

A equação (3.14) é uma representação usual na literatura de *Identificação e Controle Digital*. Os polinômios $A(z^{-1})$ e $B(z^{-1})$, relacionados aos pólos e zeros de malha aberta do processo, respectivamente, são da forma

$$A(z^{-1}) = 1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_{na} z^{-na}$$

$$B(z^{-1}) = b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_{nb} z^{-nb}$$

A correspondente equação a diferenças da planta é

$$y(t) + a_1 y(t-1) + \dots + a_{na} y(t-na) = b_0 u(t-d) + b_1 u(t-d-1) + \dots + b_{nb} u(t-d-nb)$$

onde $y(t)$ e $u(t)$ são os sinais de saída e controle, respectivamente, disponíveis nos instantes de amostragem $t = nT_s$, onde $n \in \mathbf{Z}^+$.

2.3.1 Eq. a diferenças a partir da FT discreta

Exemplo 3.4 - Obter a equação a diferenças equivalente e avaliar as características de malha aberta do modelo discreto relativo ao processo de segunda ordem representado por

$$G_p(s) = \frac{1}{(3s+1)(5s+1)}$$

Considerando o segurador de ordem zero e $T_s = 1 \text{ seg}$, então com o auxílio do *Matlab* tem-se

$$y(t) = 1.5353y(t-1) - 0.5866y(t-2) + 0.0279u(t-1) + 0.0234u(t-2)$$

De acordo com a equação (3.14) observa-se que

$$A(z^{-1}) = 1 - 1.5353z^{-1} + 0.5866z^{-2} \quad ; \quad B(z^{-1}) = 0.0279 + 0.0234z^{-1} \quad ; \quad d=1$$

Assim, como as raízes relativas aos pólos e zeros posicionam-se no interior do círculo unitário no plano complexo z o sistema é estável e de fase mínima. Os resultados do exemplo (3.4) podem ser obtidos executando-se o programa em *Matlab* da tabela (3.1).